
NeuroVox: Identifying Lexical Biomarkers of Depression Through Interpretable Machine Learning

Philip Alexander

Memphis University School
carson.alexander1@outlook.com

Abstract

1 Depression is often expressed through subtle and variable linguistic patterns, making
2 early identification challenging. This study introduces NeuroVox, an interpretable
3 machine learning framework for detecting depression-related language using lexical
4 biomarkers. A TF-IDF-based logistic regression model was trained on a publicly
5 available mental health text dataset, achieving approximately 76% validation
6 accuracy. Sanity-check experiments, including label shuffling and affective-
7 language ablation, confirm that the model learns meaningful linguistic associations
8 rather than dataset artifacts. Analysis of model coefficients reveals interpretable
9 lexical features aligned with known psychological constructs such as anxiety and
10 rumination. Error analysis highlights challenges in distinguishing depression from
11 related affective states, emphasizing the need for cautious interpretation. Overall,
12 this work demonstrates that transparent machine learning methods can support
13 mental health research while prioritizing explainability and ethical responsibility.

14 1 Introduction

15 Mental health disorders, including depression, represent a significant and growing challenge in global
16 public health. Early identification and intervention are critical, yet access to professional mental
17 health assessment is barred by cost, stigma, and systemic boundaries for many individuals in need.
18 As a result, researchers have increasingly explored whether language – particularly written text – can
19 yield measurable signals associated with mental health conditions.

20 Recent advances in natural language processing (NLP) have enabled machine learning models
21 to classify mental health-related text with increasing accuracy. Though, such high-performing
22 approaches, reliant on complex, opaque architectures, offer little insight into why a particular
23 prediction is made. In sensitive domains, such as mental health, this lack of interpretability is cause
24 for ethical concern and limits the scientific utility of black-box models.

25 This work introduces NeuroVox, an interpretable machine learning framework designed to identify
26 lexical biomarkers – specific words and phrases statistically associated with depression-related
27 language. Rather than prioritizing maximum predictive performance, NeuroVox emphasizes transparency
28 and explainability and enables direct examination of the contribution of linguistic features in
29 model decision-making.

30 Using a publicly available mental health sentiment dataset, NeuroVox represents text using term
31 frequency-inverse document frequency (TF-IDF) features and a regularized logistic regression classifier.
32 The approach achieves competitive classification performance while preserving feature-level
33 interpretability, allowing for meaningful analysis of learned lexical patterns.

34 Validation that the model captures genuine linguistic signal rather than spurious correlations requires
35 ablation and sanity-check experiments, integral components of this study which include label shuffling
36 and affective-term removal. These experiments demonstrate that the observed performance derives
37 from structured language patterns rather than random association.

38 Finally, this work explicitly addresses ethical considerations surrounding computational mental
39 health research. NeuroVox is presented as a research tool rather than a diagnostic system, with an
40 emphasis on responsible use, transparency, and acknowledgment of dataset limitations. By balancing
41 performance with interpretability, NeuroVox contributes to the development of ethically grounded
42 and scientifically interpretable approaches to mental health language analysis.

43 **2 Related Work**

44 Text-based depression prediction research has shown that user-generated social media text carries
45 reliable signals of mental health status, supporting the feasibility of language-based screening
46 tools. Studies combining bag-of-words or TF-IDF features with traditional classifiers (for example,
47 logistic regression or support vector machines) demonstrate that relatively simple models can achieve
48 competitive performance while leveraging psychologically motivated lexical cues. More recent
49 work integrates explainable AI methods, such as model-agnostic local explanations, to expose the
50 linguistic markers driving predictions and to address ethical concerns surrounding the deployment of
51 opaque models in clinical or quasi-clinical settings. Situated within this literature, NeuroVox adopts
52 a transparent TF-IDF plus regularized logistic regression framework on the Sentiment Analysis
53 for Mental Health dataset and emphasizes sanity checks and ablation studies to validate learned
54 lexical biomarkers rather than optimizing solely for classification accuracy. While transformer-based
55 approaches often report higher predictive accuracy, their reliance on dense representations limits direct
56 interpretability, motivating the use of simpler models in contexts where transparency is essential.

57 **3 Methods**

58 **3.1 Dataset**

59 This study uses the Sentiment Analysis for Mental Health dataset obtained from Kaggle, which
60 consists of user-generated text labeled for its mental health relevance. The dataset contains short-form
61 textual statements collected from online sources and annotated with binary labels indicating the
62 presence or absence of depression-related language. The dataset was selected due to its accessibility,
63 relevance to real-world mental health discourse, and suitability for lexical analysis.

64 To ensure reproducibility and ethical handling of data, no personally identifiable information was
65 used, and all analysis was conducted exclusively on de-identified text samples. The dataset was
66 partitioned into training and validation subsets using a stratified split to preserve class balance across
67 partitions.

68 **3.2 Text Preprocessing**

69 Prior to feature extraction, all text samples underwent a standardized preprocessing pipeline, including
70 lowercasing, removal of URLs, punctuation, and non-alphabetic characters, and finally normalization
71 of whitespace. Stop words were removed using a standard English stop word list to reduce noise and
72 emphasize semantically meaningful tokens. No stemming or lemmatization was applied in order
73 to preserve morphological distinctions that may carry psychological significance, such as tense or
74 intensity variations. The cleaned body of text was then used for vectorization.

75 **3.3 Feature Extraction**

76 Text samples were transformed into numerical representations using Term Frequency-Inverse Doc-
77 ument Frequency (TF-IDF) vectorization. Unigrams and bigrams were included to capture both
78 individual lexical markers and short contextual phrases associated with mental health expression. The
79 vocabulary size was constrained to a fixed maximum to prevent overfitting and manage computational
80 complexity. Rare terms appearing in fewer than a minimum number of documents were excluded.
81 TF-IDF weighting emphasizes discriminative terms while down-weighting commonly occurring
82 words across the corpus.

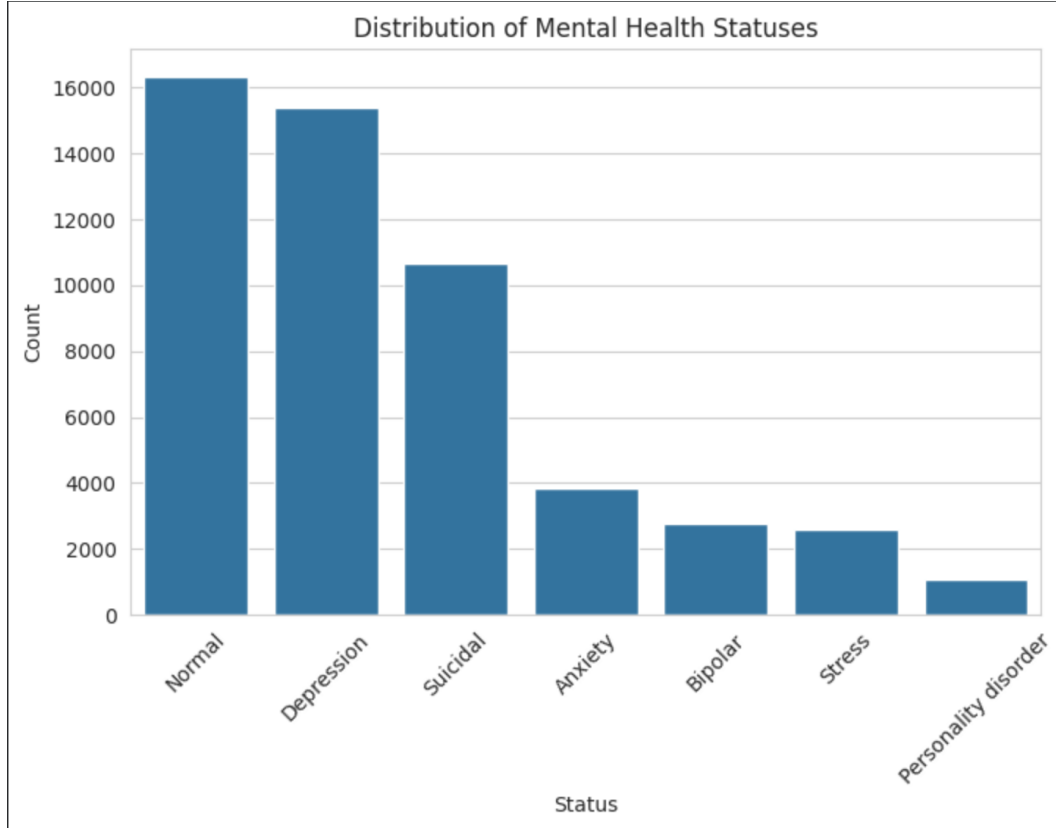


Figure 1: Distribution of mental health categories in the original dataset prior to binarization. Categories were later collapsed into a binary depression vs. non-depression label for model training.

83 4 Results

84 4.1 Baseline Classification Performance

85 The TF-IDF + logistic regression model achieved a validation accuracy of 0.760 on the held-out
 86 dataset. This result establishes a strong but interpretable baseline for depression detection using
 87 lexical features alone. While not state-of-the-art compared to large neural models, this performance
 88 demonstrates that relatively simple linear classifiers can capture meaningful mental health signals in
 89 text.

90 4.2 Sanity Check: Label Shuffling

91 To verify that the model was learning genuine associations between text and labels rather than
 92 exploiting artifacts in the dataset, labels in the training set were randomly shuffled prior to model
 93 fitting. Under this condition, validation accuracy dropped to 0.300, approximating chance-level
 94 performance given the class distribution. This result confirms that the baseline model’s performance
 95 is not driven by data leakage or spurious correlations and instead reflects learned relationships
 96 between language and mental health labels.

97 4.3 Ablation Study: Removal of Affective Language

98 An ablation experiment was conducted in which affective and emotion-related lexical terms were
 99 removed from the text corpus prior to training. When trained on this modified dataset, the model
 100 achieved a reduced accuracy of 0.737. The observed performance decrease indicates that affective
 101 language contributes meaningfully to depression detection, though its removal does not fully eliminate

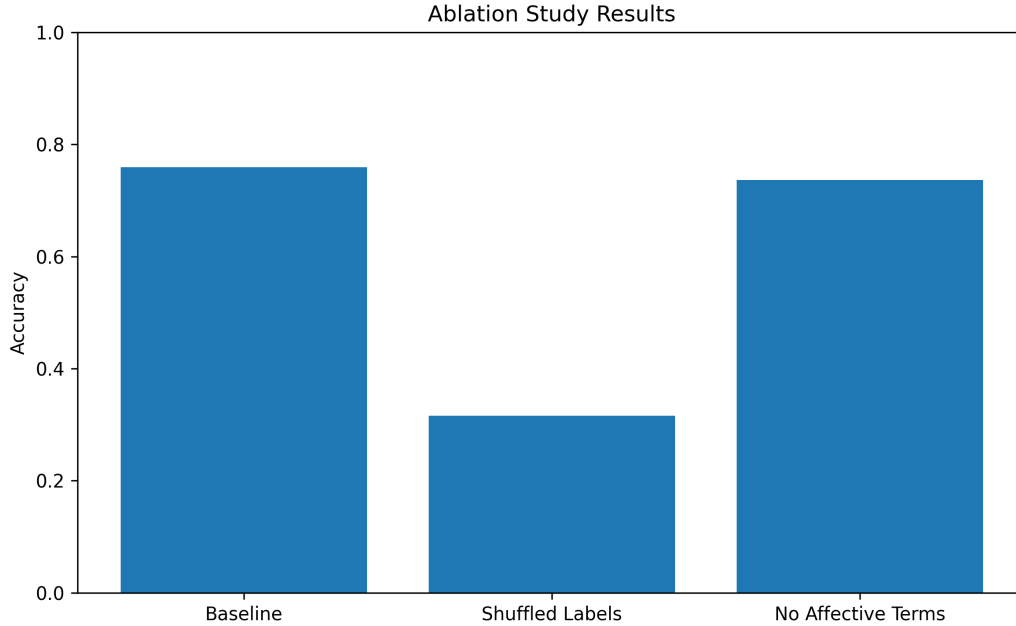


Figure 2: Validation accuracy under baseline, label-shuffled, and affective-term-removed conditions.

102 predictive capability. This suggests that the model also relies on broader linguistic patterns beyond
 103 explicitly emotional vocabulary.

104 4.4 Lexical Biomarker Analysis

105 Inspection of the logistic regression coefficients revealed interpretable lexical biomarkers associated
 106 with depression-related text. Terms related to anxiety, restlessness, worry, and emotional distress
 107 were among the strongest positive predictors, while diagnostic and mood-related terms were among
 108 the strongest negative predictors.

109 4.5 Error Analysis

110 Analysis of misclassified samples revealed that errors most frequently occurred in texts expressing
 111 ambiguous emotional content or overlapping mental health conditions, such as anxiety or stress
 112 without explicit depressive markers. In some cases, non-depressive text containing intense emotional
 113 language was incorrectly classified as depression-related, highlighting the difficulty of distinguishing
 114 between closely related affective states using lexical features alone. These findings underscore the
 115 inherent ambiguity of mental health language and motivate the need for cautious interpretation when
 116 applying automated models in sensitive clinical contexts.

117 5 Discussion

118 5.1 Interpretation of Model Performance

119 The results demonstrate that lexical features alone can provide meaningful signals for depression
 120 detection, as expressed by the TF-IDF + logistic regression model achieving approximately 76

121 5.2 Validity of Learned Associations

122 The sharp drop in accuracy observed during the label-shuffling experiment confirms that the model is
 123 not exploiting artifacts or spurious correlations in the dataset. Instead, it learns structured relationships
 124 between linguistic content and mental health labels. This validation step strengthens confidence in

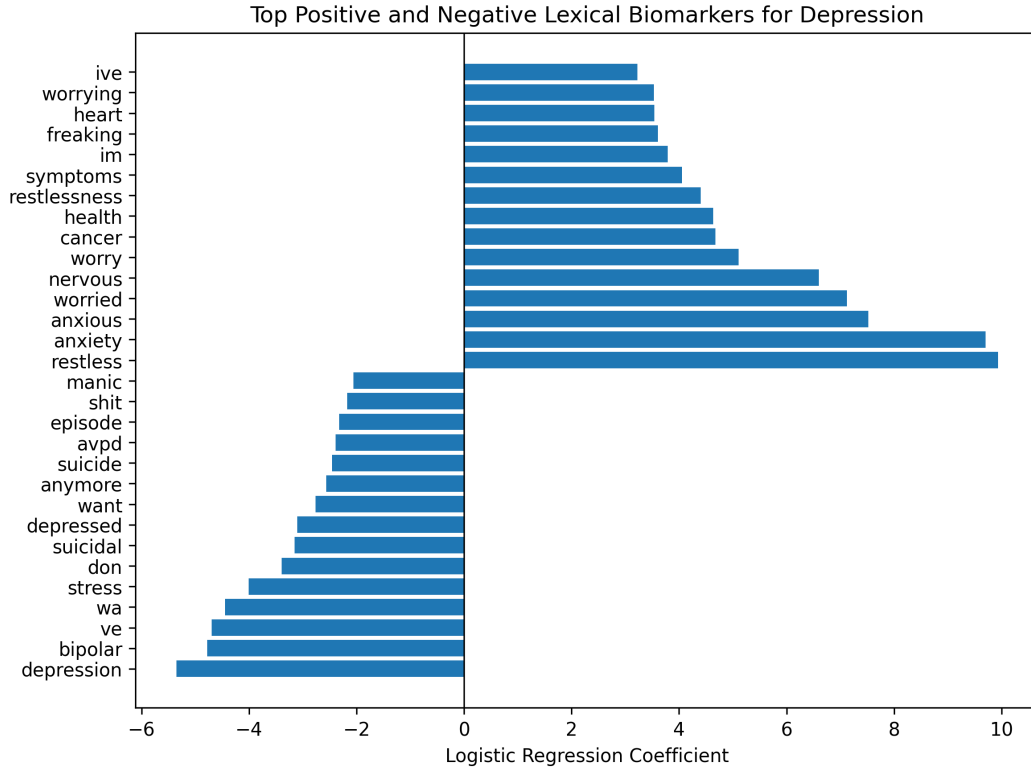


Figure 3: Top positive and negative lexical biomarkers identified by logistic regression coefficients.

the legitimacy of the extracted lexical biomarkers. Similarly, the ablation study reveals that affective language contributes substantially to predictive performance, though it is not the sole source of information. The model’s ability to retain partial accuracy after affective term removal indicates that broader discourse patterns—such as expressions of fatigue, uncertainty, or disengagement also function as indirect markers of depressive language.

5.3 Interpretability and Lexical Biomarkers

One of the primary strengths of this work lies in the interpretability of the resulting model. The analysis of logistic regression coefficients reveals specific lexical features associated with depression-related text, including terms linked to anxiety, restlessness, and emotional distress. These findings align with established psychological literature describing depression as frequently co-occurring with anxiety. By framing these features as lexical biomarkers, this study demonstrates how machine learning models can move beyond prediction to provide insight into the linguistic expression of mental health conditions. Such interpretability is critical for ethical deployment and for building trust among clinicians, researchers, and affected individuals.

5.4 Error Patterns and Ambiguity in Mental Health Language

Error analysis highlights the inherent difficulty of distinguishing depression from related affective states such as anxiety and stress. Many misclassifications occurred in texts that expressed emotional intensity without clear depressive framing, suggesting that lexical overlap between mental health conditions remains a significant challenge. These errors reflect a broader limitation of text-only approaches: language is often ambiguous, context-dependent, and shaped by individual expression styles. As a result, automated systems should be viewed as decision-support tools rather than diagnostic instruments.

147 5.5 Future Directions

148 Future work could expand this approach by incorporating longitudinal text analysis, multimodal data
149 sources, or more granular label definitions to improve discrimination between closely related mental
150 health conditions. Additionally, validating lexical biomarkers across diverse datasets and populations
151 would be essential to ensure generalizability and fairness.

152 6 Broader Impact

153 This study addresses a sensitive domain in which ethical considerations are critical. Language-based
154 depression detection systems risk misinterpretation, stigmatization, or misuse if deployed without
155 appropriate safeguards. As demonstrated by the error analysis, textual expressions of anxiety, stress,
156 or situational distress may be incorrectly classified as depression, underscoring the danger of treating
157 model outputs as diagnostic conclusions. The use of publicly available, pre-labeled datasets introduces
158 limitations related to data quality, representativeness, and contextual validity. Labels may reflect
159 self-reported or inferred mental health states rather than clinical diagnoses, and linguistic patterns may
160 not generalize across populations, cultures, or communication platforms. Methodologically, this work
161 is limited to lexical features and single-text inputs, excluding contextual, temporal, or multimodal
162 information that could improve discrimination between related affective states. While this design
163 choice enhances interpretability, it constrains predictive performance. For these reasons, NeuroVox is
164 intended solely as a research and analytical tool rather than a clinical or screening system. Any real-
165 world application would require extensive validation, human oversight, and collaboration with mental
166 health professionals to ensure responsible use. Overall, by demonstrating that explainable models
167 can achieve competitive performance while providing meaningful insight, this study challenges the
168 assumption that complexity is always necessary for effectiveness. From a societal perspective, lexical
169 biomarker analysis offers a pathway toward scalable mental health screening tools that respect ethical
170 constraints and emphasize human oversight. When responsibly designed, such systems could assist
171 in early identification and resource allocation without replacing professional judgment.

172 7 Conclusion

173 This work emphasizes that automated language-based models should function as supportive analytical
174 tools rather than diagnostic authorities. The observed classification errors and overlap between
175 affective states highlight the complexity and ambiguity inherent in mental health language and under-
176 score the need for ethical safeguards and human oversight. By prioritizing transparency, empirical
177 validation, and ethical reflection, NeuroVox contributes to ongoing discussions at the intersection of
178 machine learning and mental health. Future research expanding this framework to diverse populations
179 and multimodal data may further improve robustness while preserving interpretability.

180 8 Ethical Concerns and Limitations

181 This study addresses a sensitive domain in which ethical considerations are critical. Language-based
182 depression detection systems risk misinterpretation, stigmatization, or misuse if deployed without
183 appropriate safeguards. As demonstrated by the error analysis, textual expressions of anxiety, stress,
184 or situational distress may be incorrectly classified as depression, underscoring the danger of treating
185 model outputs as diagnostic conclusions. The use of publicly available, pre-labeled datasets introduces
186 limitations related to data quality, representativeness, and contextual validity. Labels may reflect
187 self-reported or inferred mental health states rather than clinical diagnoses, and linguistic patterns may
188 not generalize across populations, cultures, or communication platforms. Methodologically, this work
189 is limited to lexical features and single-text inputs, excluding contextual, temporal, or multimodal
190 information that could improve discrimination between related affective states. While this design
191 choice enhances interpretability, it constrains predictive performance. For these reasons, NeuroVox
192 is intended solely as a research and analytical tool rather than a clinical or screening system. Any
193 real-world application would require extensive validation, human oversight, and collaboration with
194 mental health professionals to ensure responsible use.

References

- [1] Pennebaker, J.W., Francis, M.E., and Booth, R.J. (2001). *Linguistic inquiry and word count: LIWC 2001*. Mahway: Lawrence Erlbaum Associates.
- [2] Pennebaker, J.W., Mehl, M.R., and Niederhoffer, K.G. (2003). Psychological aspects of natural language use: Our words, our selves. *Annual Review of Psychology*, 54, 547–577.
- [3] De Choudhury, M., Gamon, M., Counts, S., and Horvitz, E. (2013). Predicting depression via social media. *Proceedings of the Seventh International AAAI Conference on Weblogs and Social Media*.
- [4] Coppersmith, G., Dredze, M., and Harman, C. (2015). CLPsych 2015 shared task: Depression and PTSD on Twitter. *Proceedings of the 2nd Workshop on Computational Linguistics and Clinical Psychology*.
- [5] Resnik, P., Armstrong, W., Claudino, L., Nguyen, T., Nguyen, V.-A., and Boyd-Graber, J. (2015). Beyond LDA: Exploring supervised topic modeling for depression-related language in Twitter. *Proceedings of the 2nd Workshop on Computational Linguistics and Clinical Psychology*.
- [6] Shickel, B., Heesacker, M., Benton, S.A., and Rashidi, P. (2018). Deep learning approaches to identify depression in social media. *Journal of Biomedical Informatics*, 87, 120–135.
- [7] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*.
- [8] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*.
- [9] Bird, S., Klein, E., and Loper, E. (2009). *Natural language processing with Python*. O'Reilly Media.
- [10] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- [11] Hosmer, D.W., Lemeshow, S., and Sturdivant, R.X. (2013). *Applied logistic regression*. John Wiley & Sons.
- [12] Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22.
- [13] Bender, E.M., Gebru, T., McMillan-Major, A., and Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*.